

HE2003 Econometrics I

Tutorial 4 (Week 5) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Lecture coverage:** Lecture 4 — *Linear Regression with One Regressor 3*: further selection-bias examples (hospitals, wages, program evaluation, surveys, ratings); Least Squares Assumption 2 (i.i.d. sampling) and why simple random sampling matters; the measure of fit R^2 and its ESS/TSS decomposition; the Capital Asset Pricing Model (CAPM) and the “beta” of a stock as a running OLS application.
- **Questions:** 4 questions from Tutorial Question Set No. 4.
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; ESS explained sum of squares, SSR residual sum of squares, $TSS = ESS + SSR$; “Least Squares Assumption 1/2/3”; critical value 1.96 for a two-sided 5% test.

Question 1 (Selection bias examples; the measure of fit R^2)

Problem

- We discussed the selection bias/survivorship bias in class. Can you give some other examples of this bias?
- Explain the measure of fit R^2 . Illustrate the meaning of explained sum of squares (ESS) and total sum of squares (TSS) by using a diagram.

(Tutorial Question Set No. 4, Q1.)

Solution

Part (a)

- Investment gain on experience:** $\text{gain}_i = \beta_0 + \beta_1 \text{experience}_i + u_i$, u_i includes ability; low-ability investors exit early $\implies \text{Cov}(\text{experience}_i, u_i) > 0$, so $E[u_i | \text{experience}_i] \neq 0$ (Least Squares Assumption 1 fails) $\implies \hat{\beta}_1$ biased upward.
- Mutual-fund returns on track record:** $\text{return}_i = \beta_0 + \beta_1 \text{record}_i + u_i$, u_i includes manager skill; poor funds are dropped from databases $\implies \text{Cov}(\text{record}_i, u_i) > 0 \implies \hat{\beta}_1$ biased upward.
- Online ratings on price:** $\text{rating}_i = \beta_0 + \beta_1 \text{price}_i + u_i$, u_i is satisfaction; only extreme buyers post reviews, so the sample is selected on $u_i \implies$ polarised ratings.

Part (b)

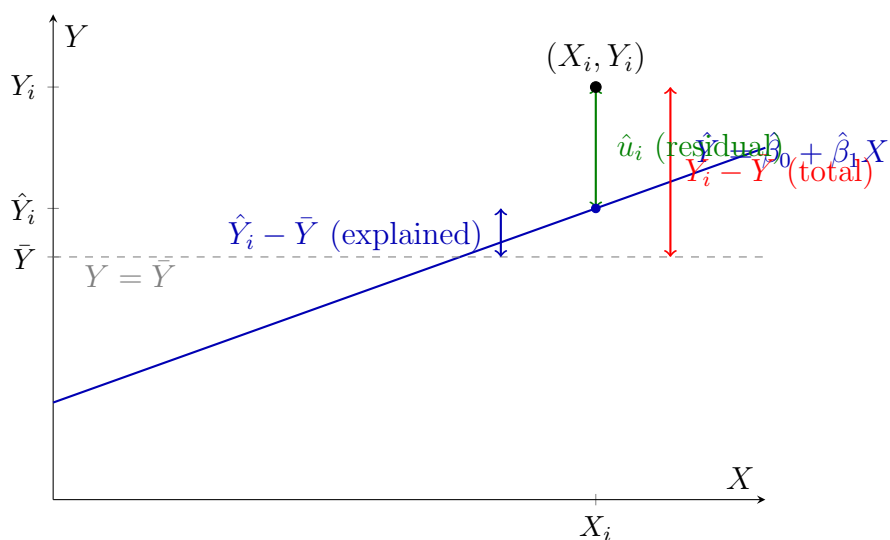
$$Y_i = \hat{Y}_i + \hat{u}_i$$

$$\text{ESS} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2, \quad \text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad \text{SSR} = \sum_{i=1}^n \hat{u}_i^2$$

$$\text{TSS} = \text{ESS} + \text{SSR} \quad \left(\sum_{i=1}^n \hat{u}_i = 0, \sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0 \right)$$

$$\text{SSR} \geq 0 \implies 0 \leq \text{ESS} \leq \text{TSS}, \quad \boxed{R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}}$$

R^2 is the fraction of the sample variation in Y explained by X , $R^2 \in [0, 1]$ ($R^2 = 0$ if $\hat{\beta}_1 = 0$; $R^2 = 1$ if every $\hat{u}_i = 0$). In the diagram, red = blue + green, with $\text{TSS} = \sum_{i=1}^n (\text{red})^2$, $\text{ESS} = \sum_{i=1}^n (\text{blue})^2$, $\text{SSR} = \sum_{i=1}^n (\text{green})^2$:



Example. $X = (1, 2, 3, 4)$, $Y = (2, 3, 3, 8)$: $\bar{X} = 2.5$, $\bar{Y} = 4$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{9}{5} = 1.8, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = -0.5$$

$$\hat{Y} = (1.3, 3.1, 4.9, 6.7)$$

$$\text{ESS} = 2(2.7^2) + 2(0.9^2) = 16.20, \quad \text{TSS} = 4 + 1 + 1 + 16 = 22.00, \quad \text{SSR} = 5.80$$

$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{16.20}{22.00} = 0.7364.$

Question 2 (Predicting weight from height)

Problem

A random sample of 100 20-year-old men is selected from a population and these men's height and weight are recorded. An OLS regression of weight on height yields

$$\widehat{\text{Weight}} = -79.24 + 4.16 \times \text{Height}, \quad (1)$$

where Weight is measured in pounds and Height is measured in inches.

- What is the regression's weight prediction for someone who is 64 inches tall? 68 inches tall? 72 inches tall?
- A man has a late growth spurt and grows 2 inches over the course of a year. What is the regression's prediction for the increase in this man's weight?

(Tutorial Question Set No. 4, Q2. No official solution is given on the Tutorial 4 slides – the “Question 2” slide is blank – so this solution is supplied in full.)

Solution

Part (a)

$$\widehat{\text{Weight}}(64) = -79.24 + 4.16(64) = 187.00$$

$$\widehat{\text{Weight}}(68) = -79.24 + 4.16(68) = 203.64$$

$$\widehat{\text{Weight}}(72) = -79.24 + 4.16(72) = 220.28$$

$\begin{aligned} \widehat{\text{Weight}}(64) &= 187.00, \\ \widehat{\text{Weight}}(68) &= 203.64, \\ \widehat{\text{Weight}}(72) &= 220.28 \text{ pounds.} \end{aligned}$

Part (b)

Method 1: substitution.

$$\widehat{\text{Weight}}(66) - \widehat{\text{Weight}}(64) = (-79.24 + 4.16(66)) - 187.00 = 195.32 - 187.00 = 8.32$$

Method 2: slope times change.

$$\Delta \widehat{\text{Weight}} = \hat{\beta}_1 \Delta H = 4.16 \times 2 \quad (\text{linear fit: intercept cancels})$$

$\Delta \widehat{\text{Weight}} = \hat{\beta}_1 \times 2 = 8.32 \text{ pounds.}$
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Question 3 (Consequences of $\hat{\beta}_1 = 0$ for $\hat{Y}_i$ and R^2)

Problem

- (a) Show that the average of the OLS predicted values $\hat{Y}_i$, $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i$, is equal to the sample average of Y , $\bar{Y}$.
- (b) Show that if the OLS estimator $\hat{\beta}_1 = 0$, then $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$.
- (c) Show that if (b) holds, then $R^2 = 0$.

(Tutorial Question Set No. 4, Q3.)

Solution

Part (a)

Method 1: residuals.

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i + \sum_{i=1}^n \hat{u}_i = \sum_{i=1}^n \hat{Y}_i \quad \left(\sum_{i=1}^n \hat{u}_i = 0, \text{ FOC} \right)$$

Method 2: intercept formula.

$$\sum_{i=1}^n \hat{Y}_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n X_i = n(\hat{\beta}_0 + \hat{\beta}_1 \bar{X}) = n\bar{Y} \quad (\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X})$$

$$\boxed{\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}.}$$

Part (b)

$$\begin{aligned} \hat{\beta}_1 = 0 &\implies \hat{Y}_i = \hat{\beta}_0 + 0 \cdot X_i = \hat{\beta}_0 \\ \hat{\beta}_0 &= \bar{Y} - 0 \cdot \bar{X} = \bar{Y} \end{aligned} \quad (\text{OLS intercept})$$

$$\boxed{\hat{Y}_i = \hat{\beta}_0 = \bar{Y} \quad \text{for every } i = 1, \dots, n.}$$

Part (c)

Method 1: ESS = 0.

$$\text{ESS} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n (\bar{Y} - \bar{Y})^2 = 0 \quad (\text{part (b)})$$

Method 2: $SSR = TSS$.

$$\hat{u}_i = Y_i - \hat{\beta}_0 - 0 \cdot X_i = Y_i - \bar{Y}$$

$$SSR = \sum_{i=1}^n (Y_i - \bar{Y})^2 = TSS \implies ESS = TSS - SSR = 0 \quad (TSS = ESS + SSR)$$

$$R^2 = \frac{ESS}{TSS} = \frac{0}{TSS} = 0 \quad (\text{provided } TSS > 0).$$

Question 4 (CAPM: comparing assets with equal β but different α)

Problem

As explained in class, we can use OLS to estimate the Capital Asset Pricing Model (CAPM). The linear regression model is defined as

$$(R - R_f) = \alpha + \beta(R_m - R_f) + u, \quad (2)$$

where $R - R_f$ denotes the excess return of the stock (or hedge fund), $R_m - R_f$ denotes the excess return of the market (e.g., R_m can be the return of S&P 500), and R_f denotes the return of the risk-free asset (e.g., US government bond). In practice, we report the OLS estimates $\hat{\alpha}$ and $\hat{\beta}$ for the stock (or fund). Under the standard CAPM theory, $\alpha = 0$. Therefore, if we find $\hat{\alpha}$ is significantly positive for a certain stock (or fund), then it is especially valuable as it can beat the prediction from CAPM (i.e., it has a premium that cannot be explained by CAPM). Compare the return and risk of a stock (or a fund) with different values of α but the same value of β . What do you find? [Hint: find its expected value and variance.] (*Tutorial Question Set No. 4, Q4. No official solution is given on the Tutorial 4 slides – both “Question 4” slides in the deck are blank – so this solution is supplied in full, extending the Lecture 4 CAPM material.*)

Solution

Let $Y = R - R_f$, $X = R_m - R_f$. Assumption 1: $E[u | X] = 0 \implies E[u] = 0$, $\text{Cov}(X, u) = 0$ (LIE).

$$\begin{aligned} E[R - R_f] &= \alpha + \beta E[R_m - R_f] & (E[u] = 0) \\ E[R_A - R_f] - E[R_B - R_f] &= \alpha_A - \alpha_B > 0 & (\beta_A = \beta_B, \alpha_A > \alpha_B) \\ \text{Var}(R - R_f) &= \beta^2 \text{Var}(X) + \text{Var}(u) + 2\beta \text{Cov}(X, u) & (\alpha, \beta \text{ constants}) \\ &= \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u) & (\text{Cov}(X, u) = 0) \end{aligned}$$

$$\boxed{\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u),}$$

$\therefore \alpha$ does not enter Var: higher α gives higher expected excess return at the same risk (same β , same $\text{Var}(u)$).

Illustration (monthly, both funds): $E[R_m - R_f] = 0.6\%$, $\text{sd}(R_m - R_f) = 4\%$, $\beta = 1.2$, $\text{sd}(u) = 2\%$; $\alpha_A = 0$, $\alpha_B = 0.3\%$.

$$\text{Var}(R - R_f) = (1.2)^2(0.04)^2 + 0.0004 = 0.002704, \quad \text{sd} = 5.20\% \quad (\text{both funds})$$

$$E[R_A - R_f] = 0 + 1.2(0.6\%) = 0.72\%, \quad E[R_B - R_f] = 0.3\% + 1.2(0.6\%) = 1.02\%$$

$$\boxed{\text{Fund } B \text{ earns } 0.30 \text{ pp more per month, at the identical risk } \text{sd} = 5.20\%.$$

Final Answers

- 1(a) Investment gain on experience; mutual-fund returns on track record; online ratings on price.
- 1(b) $R^2 = \text{ESS}/\text{TSS} = 1 - \text{SSR}/\text{TSS}$; example: $\text{TSS} = 22.00$, $\text{ESS} = 16.20$, $\text{SSR} = 5.80$, $R^2 = 0.7364$.
- 2(a) $\widehat{\text{Weight}}(64) = 187.00$, $\widehat{\text{Weight}}(68) = 203.64$, $\widehat{\text{Weight}}(72) = 220.28$ pounds.
- 2(b) $\Delta \widehat{\text{Weight}} = \hat{\beta}_1 \times 2 = 8.32$ pounds.
- 3(a) $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}$.
- 3(b) $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$ for every i .
- 3(c) $\text{ESS} = 0 \Rightarrow R^2 = 0$ ($\text{TSS} > 0$).
- 4 $E[R - R_f] = \alpha + \beta E[R_m - R_f]$; $\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u)$; Fund B earns 0.30 pp more per month at identical sd = 5.20%.